

Mini Cases

Risk Management

2025/2026

Please upload your solutions and indicate which Mini Cases (MCs) you have solved before **May 20, 2026 at 7am**. To do so, go to the assignment section on Canvas and choose “Unit 1 Indication and Submission of MCs”. Then answer the true/false questions (Questions 1 to 6) and upload your solutions file (Question 7). **There will be no deadline extensions! Only complete submissions containing answers to the T/F questions and a solutions upload will be considered, your last complete attempt counts!** You can either write your solutions using software or scan your handwritten solutions. Please note that you may only submit one solutions file. If your solutions consists of more than one file (e.g. 1 pdf and 1 Excel file), please upload a .zip file. Uploaded files will be checked for plagiarism. The solutions in your file should be self-explanatory - not only for the instructor, but for other students as well. Indicate only Mini Cases you feel confident to present.

Important remark for the presentations: Make sure to provide a concise explanation of your solutions. In particular, when applying any formula from the lecture materials or other sources, you may be asked to explain the structure of your solutions and provide the intuition and underlying concepts behind the formula used.

Mini Case 2.1.

Consider a risk-neutral investor who wants to buy a bond that defaults with a probability of 15% and in case of a default, it would be possible to recover 35%. The risk free rate is equal to 3%.

- What is the minimum nominal interest rate such that the investor would actually buy the bond?
- Assume there is another investor who is not perfectly diversified and risk averse. Would he also buy the bond at this interest rate?
- Next, assume that the LGD is lower and only equal to 0.55. Does this increase or decrease the minimum nominal interest rate?

Mini Case 2.2.

Consider a risk-neutral investor and three bonds with an LGD of 50%, 75%, and 100%, respectively. The risk-free rate is equal to 2.5%. Next, assume that you are uncertain about the PD of the individual bonds and each of the values in "Mini_Case_2.2.xlsx" or "Mini_Case_2.2.Rdata" is possible.

- Calculate the required minimum interest rate for each bond and PD combination.
- Plot all values in one graph and discuss the results.

Mini Case 2.3.

Consider a firm with assets A worth 300 and asset volatility 4%. The firm's target rating is BBB.

Rating Scale	1Y PD
AAA	0.00%
AA	0.01%
A	0.28%
BBB	1.16%
BB	4.36%
B	18.09%
C	42.95%

- Determine the firm's equity such that the 1-year PD matches the target rating.
- Assume that the value of equity is 33.24. Which credit rating is the firm assigned?
- Assume that the value of equity is 33.24, however the asset volatility is equal to 6.48%. What is the probability of default?

Mini Case 2.4.

A firm operates in two business lines, A and B, with the following characteristics.

Asset	Value	Volatility	Expected Return (EUR)
A	150	10%	7.5
B	100	15%	2.5

Based on historical information, the firm assumed that asset returns are normally distributed with the given parameters. In addition, the firm assumed that asset returns are uncorrelated.

- Compute the 1-year RORAC (at the 2.5%-level) of the firm given this base case information.
- Given the Covid-19-crisis and the turmoil in the energy markets, the firm assumes that asset returns might be markedly correlated in the future. At the same time, however, the owners of the firm decided that they would be

satisfied with a lower RORAC of just 25% in the future. Compute the critical value of asset return correlation that must not be exceeded to reach the 25% RORAC goal.

Mini Case 2.5.

Consider a firm with a target rating that corresponds to a PD of 4.35%.

- Compute the minimum expected rate of return such that the RORAC is greater than 10% assuming normally distributed returns with a volatility of 25%.
- How does this minimum rate change if the firm is upgraded, i.e. its PD equals 1.16%?
- In a pure Modigliani-Miller setup (perfect market) the firm's capital structure decision and the asset risk profile are irrelevant. Discuss if this is applicable in practice.

Mini Case 2.6.

Your research department provided you with a sample of possible states that the stock market and two investment projects can be in one year from now (see "Mini_Case_2.6.xlsx" or "Mini_Case_2.6.Rdata"). Assume that all states are equally likely. Today both investment projects would require an initial investment of $I_{P_{1,2}} = 150$ and the total market has a value of 1000. In which project would you invest if ...

- you decide based on ROI?
- the risk free rate is equal to 0.04 and you want to maximize the Sharpe Ratio?
- you want to maximize RORAC based on a 1% VaR?
- you want to maximize RORAC based on a 1% VaR of systematic risk?

Hint:

*For project i the VaR based on systematic risk is given by the respective quantile of $\beta_i * r_m * I_{P_i}$, where r_m denotes the market return.*